

# DMSC: Dynamic Multi-Scale Coordination Framework for Time Series Forecasting

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## Abstract

Time Series Forecasting (TSF) faces persistent challenges in modeling intricate temporal dependencies across different scales. Despite recent advances leveraging different decomposition operations and novel architectures based on CNN, MLP or Transformer, existing methods still struggle with static decomposition strategies, fragmented dependency modeling, and inflexible fusion mechanisms, limiting their ability to model intricate temporal dependencies. To explicitly solve the mentioned three problems respectively, we propose a novel Dynamic Multi-Scale Coordination Framework (DMSC) with Multi-Scale Patch Decomposition block (EMPD), Triad Interaction Block (TIB) and Adaptive Scale Routing MoE block (ASR-MoE). Specifically, EMPD is designed as a built-in component to dynamically segment sequences into hierarchical patches with exponentially scaled granularities, eliminating predefined scale constraints through input-adaptive patch adjustment. TIB then jointly models intra-patch, inter-patch, and cross-variable dependencies within each layer’s decomposed representations. EMPD and TIB are jointly integrated into layers forming a multi-layer progressive cascade architecture, where coarse-grained representations from earlier layers adaptively guide fine-grained feature extraction in subsequent layers via gated pathways. And ASR-MoE dynamically fuses multi-scale predictions by leveraging specialized global and local experts with temporal-aware weighting. Comprehensive experiments on thirteen real-world benchmarks demonstrate that DMSC consistently maintains state-of-the-art (SOTA) performance and superior computational efficiency for TSF tasks. Code is available at <https://github.com/1327679995/DMSC>.

## Introduction

Time Series Forecasting (TSF) constitutes a pivotal capability across numerous fields such as energy consumption(Alvarez et al. 2010; Guo et al. 2015), healthcare monitoring(Tran, Nguyen, and Shahabi 2019; Wei et al. 2022), transportation scheduling(Guo et al. 2020; Jin et al. 2021), weather forecasting(Wu et al. 2023b; Bi et al. 2023), economics(Chen et al. 2023b; Yu et al. 2023) and urban computing(Li et al. 2024; Yuan et al. 2024). Time series data exhibit intricate temporal dependencies, encompassing non-linear relationships, long-term and short-term patterns such as trends, seasonality and dynamic variable couplings(Qiu

et al. 2024). Owing to non-stationarity(Nason 2006) and complex interconnections(Dagum and Bianconcini 2016), traditional approaches struggle to effectively deal with time series data, making effectively dependency modeling a critical challenge for enhancing predictive precision(Shao et al. 2025). In recent years, the powerful ability of deep learning to capture complex dependency relationships in time series data has been demonstrated, making it a formidable tool in the field of TSF(Chen et al. 2023a).

However, intricate dependencies in time series inherently manifest across multiscales: coarser granularities encapsulate long-term trends while finer resolutions capture short-term fluctuations and periodic patterns. Existing single-scale modeling approaches often fail to balance fine-grained local details and holistic global trends(Wang et al. 2025). To effectively model multiscale temporal dependencies, recent research has pioneered multiscale modeling frameworks that concurrently extract global and local patterns, thereby mitigating the representational limitations of single-scale approaches. While previous implementations often employ moving average downsampling(Wang et al. 2024b; Huang et al. 2025), which often remains insufficient in capturing local semantic information. In contrast, more effective alternatives focus on patch-wise and variable-wise decomposition operations that decompose sequence across feature and temporal dimensions. Such decomposition operations better preserves both global trends and local semantics, and multi-length patches inherently learn multi-scale representations. Critically, prevailing methods exhibit two constraints:1) dependence on fixed-scale decomposition strategies that lack dynamic scale adaptation; 2) fragmented processing of temporal dependencies and cross-variable interactions, undermining holistic dependency modeling.

The core value of multiscale feature extraction resides in mining complementary information across diverse temporal granularities and hierarchical levels, thereby overcoming the limitations of single-scale modeling. Following feature capture, the effective design of prediction heads for multiscale fusion is equally important. Conventional approaches typically employ either linear projection layers(Li, Li, and Diao 2025), additive combinations of multiple projections(Wang et al. 2024b), or spectral amplitude-weighted aggregation(Wu et al. 2023a). However, above simplistic fusion method exhibit fundamental inadequacies: they fail

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to leverage heterogeneous dominance patterns across multi-scales, disregard dynamic inter-scale importance, and induce quadratic parameter growth with increasing input lengths and scale counts, severely compromising deployment efficiency(Qiu et al. 2024).

To effectively modeling complex temporal dependencies which inherently manifest across multiple patterns and scales, we propose a Dynamic Multi-Scale Coordination framework (DMSC) for TSF. This framework dynamically decomposes time series and extracts features across scales. It effectively captures intricate dependencies across temporal resolutions and cross-variable interactions. Meanwhile, it can adaptively predict learned representations through temporal-aware fusion. Specifically, DMSC processes intricate dependencies through a multi-layer progressive architecture and Adaptive Scale Routing Mixture-of-Experts (ASR-MoE). The multi-layer progressive cascade architecture incorporates Embedded Multi-Scale Patch Decomposition Block (EMPD), Triad Interaction Block (TIB), where EMPD transforms input into 3D representations at varying granularities and TIB models heterogeneous dependencies within each layer. In this architecture, coarse-grained representations from shallow layers adaptively guide fine-grained feature extraction in subsequent layers via gated pathways. After multi-scale feature extraction, ASR-MoE is employed to perform time-aware weighted dynamic fusion for multi-scale prediction, enabling adaptive integration of features across scales. By synergistically integrating dynamic patch decomposition, deeper models and sparse MoE principles, DMSC achieves full-spectrum multi-scale coordination across embedding, feature extraction, and prediction stages. This dynamic co-optimization enable it to achieve SOTA performance across 13 real-world benchmarks while maintaining high efficiency and low computational complexity. Our contributions are as follows:

- A novel multi-layer progressive cascade architecture is designed to jointly integrate Embedded Multi-Scale Patch Decomposition Block (EMPD) and Triad Interaction Block (TIB) across layers. Unlike fixed-scale approaches, EMPD dynamically adjusts patch granularities by employing a lightweight neural controller based on input characteristics, and TIB jointly models intra-patch, inter-patch, and inter-variable dependencies through gated feature fusion to form a coarse-to-fine feature pyramid, where coarse-grained representations from earlier layers adaptively guide fine-grained extraction in subsequent layers via gated residual pathways.
- Adaptive Scale Routing Mixture-of-Experts (ASR-MoE) is proposed to resolve static fusion limitations in prediction. ASR-MoE establishes a hierarchical expert architecture to explicitly decouples long-term and short-term dependencies, where global-shared experts capture common long-term temporal dependencies, while local-specialized experts model different short-term variations. Significantly, a temporal-aware weighting aggregator is designed to dynamically compute scale-specific prediction contributions with historical memory.
- Extensive experiments demonstrate that DMSC achieves

SOTA performance and superior efficiency across multiple TSF benchmarks.

## Related Work

### Time Series Forecasting Models

Existing deep models for TSF tasks can be broadly categorized into MLP-based, CNN-based, and Transformer-based architectures(Wang et al. 2024c). MLP-based models(Oreshkin et al. 2020; Yi et al. 2023; Chi et al. 2021) primarily model temporal dependencies through predefined decomposition schemes and fully-connected architectures. While demonstrating simplicity and efficiency in capturing intra-series patterns, they exhibit critical limitations in modeling long-term dependencies and complex inter-series relationships. In contrast, CNN-based models(Bai, Kolter, and Koltun 2018; LIU et al. 2022; Wang et al. 2023) leverage different convolutions to effectively capture local dependencies. Although they remain constrained in modeling global relationships, recent efforts have mitigated this through large-kernel convolutions(Luo and Wang 2024) and frequency-domain analysis(Wu et al. 2023a). Nevertheless, Transformer-based models(Zhou et al. 2021; Wu et al. 2021; Wang et al. 2024a) have emerged as a powerful paradigm by utilizing self-attention mechanisms(Vaswani et al. 2017) to model superior long-term dependency across long sequences, so that effectively capturing persistent temporal relationships. However, Transformer-based models face substantive criticism regarding permutation invariance and quadratic computational complexity limitations in recent researches(Zeng et al. 2023; Tang and Zhang 2024). While significant progress has been made, existing deep architectures primarily operate on single or fixed decomposition, and often model dependencies in a fragmented manner, lacking a mechanism for dynamic, adaptive multi-scale decomposition and coordinated dependency modeling.

### Decomposition Operations in TSF

To effectively address intricate temporal dependencies, most studies have adopted multiscale modeling perspective, employing diverse multiscale decomposition operations on time series. TimesNet(Wu et al. 2023a) learns frequency-adaptive periods in the frequency domain, transforming time series into 2D tensors to explicitly model intra-period and inter-period variations. TimeMixer(Wang et al. 2024b) decomposes sequences into seasonal and trend components(Wu et al. 2021), subsequently applying different granularity mixing strategies to integrate multiscale features. However, simply using moving average inadequately preserves local semantic patterns. Diverging from the above operations, some methods focus on variable-wise and patch-wise disentanglement. PatchTST(Nie et al. 2023) segments series into subseries-level patches as input tokens, employing channel independence to capture local semantics. ITransformer(Liu et al. 2024) embeds entire series as singular tokens to model extended global representations. TimeXer(Wang et al. 2024d) hierarchically represents variables through dual disentanglement, variable-wise for cross-channel interactions and patch-wise for endogenous vari-

ables. However, these decomposition techniques are constrained by static or predefined scale configurations, which hinders their adaptability to complex temporal patterns.

### Multiscale Prediction Operations in TSF

Meanwhile, researches also focuses on multiscale operations in prediction heads. TimeMixer(Wang et al. 2024b) employs dedicated projectors-one per scale-to generate scale-specific predictions. Diverging from simplistic additive aggregation, TimeMixer++(Wang et al. 2025) utilizes frequency-domain amplitude analysis to weightedly aggregate projector outputs. TimeKAN(Huang et al. 2025) adopts frequency transformation and padding operations to unify multiscale representations into identical dimensions, enabling holistic scale integration. Current multiscale fusion mechanisms predominantly rely on static weighting schemes or fixed designs. They fail to dynamically prioritize different scales with intricate temporal dependencies. In contrast to these fragmented approaches, DMSC pioneers fully-spectrum multiscale coordination spanning embedding, extraction, and prediction stages, achieving optimal balance between computational efficiency and forecasting accuracy.

### Dynamic Multi-Scale Coordination Framework

Time series forecasting addresses the fundamental challenge of predicting future values  $Y = y_{t+1}, \dots, y_{t+\tau} \in \mathbf{R}^{\tau \times C}$  from historical observations  $X = x_1, \dots, x_t \in \mathbf{R}^{t \times C}$ , where  $\tau$  is the prediction horizon,  $t$  is the history horizon, and  $C$  is the number of variate. To model complex multiscale dependencies inherent in real-world series, the Dynamic Multi-Scale Coordination (DMSC) framework is proposed. As illustrated in Fig. 1, DMSC establishes full-spectrum multiscale coordination across embedding, feature extraction, and prediction stages using EMPD, TIB and ASR-MoE.

### Multi-Layer Progressive Cascade Architecture

The multi-layer progressive cascade architecture enables hierarchical feature learning through stacked EMPD-TIB units, creating a dynamic feature pyramid where coarse-grained representations from earlier layers adaptively guide fine-grained extraction in subsequent layers via gated pathways. Formally, given input  $\mathbf{X} \in \mathbf{R}^{C \times L}$ , layer  $l$  processes:

$$\begin{aligned} \mathcal{T} &= \{\mathbf{F}_1, \mathbf{F}_2, \dots, \mathbf{F}_l\}, \\ \mathbf{F}_l &= \mathbf{TIB}_l(\mathbf{Z}_l), \\ \mathbf{Z}_l &= \mathbf{EMPD}_l(\mathcal{G}_l(\mathbf{F}_{l-1}) + \mathbf{X}), \end{aligned} \quad (1)$$

where  $\mathcal{T}$  is the set of hierarchical features generated by this architecture for ASR-MoE, with each  $\mathbf{F}_l \in \mathbf{R}^{C \times D}$  representing the  $l$ -th layer’s output.  $\mathcal{G}_l$  is a gated projection matrix that adaptively modulates the residual flow. This cascade flow enables progressive refinement of multiscale representations, where  $\mathbf{F}_{l-1}$  adaptively guides  $\mathbf{Z}_l$  with residual connections, and  $\mathbf{TIB}_l$  iteratively enhances representations by jointly integrating intra-patch, inter-patch, and cross-variable dependencies. By interleaving dynamic patch

decomposition (EMPD) with triadic dependency modeling (TIB), our cascade architecture achieves input-aware progressive refinement of temporal granularities, enabling coherent modeling of both short-term dynamics and long-range trends while maintaining parameter efficiency.

### Embedded Multi-Scale Patch Decomposition

Patch decomposition serves as a fundamental operation to transform time series into structured representations that preserve local semantics while enabling hierarchical pattern discovery. Unlike conventional single-fixed or fixed-scale approaches, the EMPD block introduces input-adaptive and hierarchical decomposition mechanism as a built-in component, dynamically adjusting patch granularities based on sequence characteristics. Thus eliminating rigid predefined patch configurations while natively integrating multiscale decomposition into the model architecture.

Specifically, given an input time series tensor  $\mathbf{X} \in \mathbf{R}^{C \times L}$  (where  $C$  is the number of features,  $L$  is the sequence length), EMPD first computes a scale factor  $\alpha$  through a lightweight neural controller:

$$\alpha = \mathcal{N}_\theta(\Phi_{\text{GAP}}(\mathbf{X})) \quad (2)$$

where  $\mathcal{N}_\theta$  is a micro-MLP with sigmoid activation,  $\Phi_{\text{GAP}}$  compresses temporal dimensions through averaging. Factor  $\alpha \in [0, 1]$  dynamically determines the base patch length, and then the patch length decays exponentially as follows:

$$\begin{aligned} \mathbf{P}_{\text{base}} &= [\mathbf{P}_{\text{min}} + \alpha \cdot (\mathbf{P}_{\text{max}} - \mathbf{P}_{\text{min}})], \\ \mathbf{P}_l &= \max(\mathbf{P}_{\text{min}}, [\mathbf{P}_{\text{base}} / \tau^l]), \end{aligned} \quad (3)$$

where  $\mathbf{P}_{\text{max}}$  and  $\mathbf{P}_{\text{min}}$  denoting the learned parameter for minimum and maximum patch bounds, and  $\mathbf{P}_l$  means patch length in layer  $l$ , which ensures shallow layers process coarse-grained dependencies while deeper layers capture fine-grained patterns. Then EMPD applies replication padding to handle boundary effects, followed by hierarchical patch unfolding. The operations are formalized as:

$$\mathbf{X}_p^l = \text{Unfold}(\text{Padding}_{S_l}(\mathbf{X})), \quad (4)$$

where  $\text{Padding}_{S_l}(\cdot)$  denotes replication padding with stride  $S_l = P_l/2$  and  $\text{Unfold}_{P_l, S_l}$  unfolds the padded sequence into patches of length  $P_l$  with stride  $S_l$ . This operation yields a 3D patch tensor  $\mathbf{X}_u^l \in \mathbf{R}^{C \times N_l \times P_l}$ . Then EMPD transforms  $\mathbf{X}_u^l$  into a unified embedding space via linear projection:

$$\mathbf{Z}^l = \text{Projector}(\mathbf{X}_p^l), \quad (5)$$

This direct linear projection avoids redundant flattening operations, efficiently transforming patch-wise temporal features into a unified embedding space while preserving the hierarchical structure of multiscale patches.

### Triad Interaction Block

TIB is designed to model heterogeneous dependencies within the multiscale patch representations generated by EMPD, through joint capture of intra-patch, inter-patch, and cross-variable features. Given the input tensor  $\mathbf{Z}^l \in$

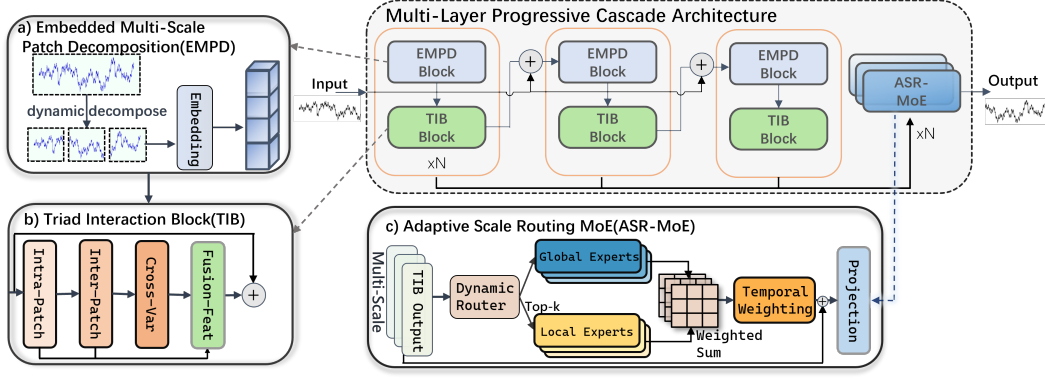


Figure 1: The framework of DMSC. EMPD dynamically segments input series into hierarchical patches, TIB jointly models three dependencies, and ASR-MoE fuses multiscale predictions adaptively.

$R^{C \times N_l \times D}$ , TIB integrates three complementary dependency types in a coherent framework. Specifically, TIB first processes intra-patch dependencies using depth-wise separable convolutions via  $\zeta_{intra}$  to capture fine-grained local patterns:

$$\mathbf{F}_{intra}^l = \zeta_{intra}(\mathbf{Z}^l), \quad (6)$$

where  $\zeta_{intra}$  consists the depth-wise convolution and Conv1d to captures local temporal contexts and project features back to the embedding space. This operation preserves intra-patch continuity while extracting localized representations.

For inter-patch dynamics  $\mathbf{F}_{intra}^l$ , TIB employs dilated convolutions with adaptive pooling via  $\zeta_{inter}$ :

$$\mathbf{F}_{inter}^l = \zeta_{inter}(\mathbf{F}_{intra}^l), \quad (7)$$

where  $\zeta_{inter}$  consists the dilated convolution and adaptive pooling to capture broader temporal contexts and patch-level information without increasing computational cost.

Then TIB models dependencies between different features by generating adaptive weighting coefficients  $\mathbf{G}^l \in \mathbf{R}^{C \times 1}$  from globally aggregated inter-patch features  $\bar{\mathbf{F}}_{inter}^l \in \mathbf{R}^{1 \times D}$  as follows:

$$\mathbf{G}^l = \sigma(\text{MLP}(\bar{\mathbf{F}}_{inter}^l)), \quad (8)$$

$$\mathbf{F}_{cross}^l = \mathbf{G}^l \odot \mathbf{F}_{inter}^l,$$

where  $\bar{\mathbf{F}}_{inter}^l$  denotes the global average of  $\mathbf{F}_{inter}^l$ , and the sigmoid function  $\sigma$  act as a feature-specific gate. The weights  $\mathbf{G}^l$  are then applied element-wise to  $\mathbf{F}_{inter}^l$  through multiple operation  $\odot$ , which adaptively scales each feature's contribution based on its own global relevance, enabling context-aware cross-variable interactions.

Finally, three dependency representations are adaptively fused via learned gating weights, and the fused features are then integrated with residual connection and normalization:

$$\mathbf{F}_{fused}^l = \sum_{i=1}^3 \mathbf{G}_i^l \odot \mathbf{F}_i^l, \quad (9)$$

$$\mathbf{F}_{output}^l = \text{LayerNorm}(\mathbf{F}_{fused}^l + \mathbf{Z}^l),$$

where  $\mathbf{G}_i^l$  is learned gating weights for each representations. This architecture enables TIB to dynamically balance contributions of different dependency types based on input characteristics, thus effectively capturing intricate dependencies, providing rich temporal features for multiscale prediction.

### Adaptive Scale Routing Mixture-of-Experts

To address the limitations of static fusion mechanisms in multiscale forecasting, we propose the Adaptive Scale Routing Mixture-of-Experts (ASR-MoE) block. This dynamic prediction head models temporal dependencies at different horizons via hierarchical experts (global-shared and local-specialized experts) and adaptively weights scale-specific predictions based on temporal patterns.

Specifically, ASR-MoE comprises two expert groups to handle distinct temporal granularities. Global experts  $\mathcal{E}^g = \{\mathcal{G}_1, \mathcal{G}_2, \dots, \mathcal{G}_m\}$  capture common long-term dependencies through deep nonlinear transformations, modeling persistent trends and long-periodic patterns that commonly existed in time series. Local experts  $\mathcal{E}^l = \{\mathcal{L}_1, \mathcal{L}_2, \dots, \mathcal{L}_n\}$  specialize in different short-term variations through shallower networks, detecting short-periodic patterns and high-frequency fluctuations that dynamically varied across stages. This explicit decoupling enables specialized handling of complex dynamics while maintaining higher parameter efficiency.

For weights calculation of each scale's experts, a dynamic routing mechanism is proposed to assign input-dependent weights to experts, balancing global and local dependencies. Given a scale-specific feature  $\mathbf{F}_l$ , the router generates weights as follows:

$$\mathbf{\Omega}_l = \text{Softmax}(\text{MLP}(\mathbf{F}_l)), \quad (10)$$

where the weight matrix  $\mathbf{\Omega}$  are divided into global weights  $\mathbf{\Omega}^G$  and local weights  $\mathbf{\Omega}^L$ . For computational efficiency, we activate only the top- $K$  local experts via sparse routing:

$$\hat{\mathbf{\Omega}}_l^L, \mathcal{I} = \text{Top-K}(\mathbf{\Omega}_l^L, K), \quad (11)$$

where  $\mathcal{I}$  denotes indices of selected experts.

Finally, a temporal-aware weighting module fuses outputs from all scales by historical scale importance. Scale-specific

| Models      | DMSC(Ours)   |              | TimeMixer    |              | iTransformer |              | PatchTST |       | Dlinear      |       | TimesNet |       | Autoformer |       | TimeXer      |              | PatchMLP |       | TimeKAN      |              | AMD   |              |
|-------------|--------------|--------------|--------------|--------------|--------------|--------------|----------|-------|--------------|-------|----------|-------|------------|-------|--------------|--------------|----------|-------|--------------|--------------|-------|--------------|
| Metric      | MSE          | MAE          | MSE          | MAE          | MSE          | MAE          | MSE      | MAE   | MSE          | MAE   | MSE      | MAE   | MSE        | MAE   | MSE          | MAE          | MSE      | MAE   | MSE          | MAE          | MSE   | MAE          |
| ETTh1       | <b>0.415</b> | <b>0.426</b> | 0.456        | 0.444        | 0.452        | 0.446        | 0.445    | 0.445 | 0.461        | 0.457 | 0.479    | 0.466 | 0.670      | 0.564 | 0.460        | 0.452        | 0.458    | 0.445 | <u>0.429</u> | <u>0.432</u> | 0.452 | 0.439        |
| ETTh2       | <b>0.355</b> | <b>0.383</b> | <u>0.372</u> | <u>0.400</u> | 0.384        | 0.407        | 0.383    | 0.412 | 0.563        | 0.519 | 0.411    | 0.421 | 0.488      | 0.494 | 0.374        | 0.404        | 0.406    | 0.422 | 0.389        | 0.409        | 0.616 | 0.558        |
| ETTh1       | <b>0.368</b> | <b>0.383</b> | 0.384        | 0.398        | 0.408        | 0.412        | 0.386    | 0.400 | 0.404        | 0.408 | 0.418    | 0.418 | 0.636      | 0.534 | 0.391        | 0.395        | 0.387    | 0.398 | <u>0.376</u> | 0.397        | 0.394 | <u>0.396</u> |
| ETTh2       | <b>0.268</b> | <b>0.317</b> | 0.278        | 0.324        | 0.292        | 0.336        | 0.288    | 0.334 | 0.354        | 0.402 | 0.291    | 0.330 | 0.502      | 0.452 | <u>0.277</u> | <u>0.323</u> | 0.287    | 0.329 | 0.282        | 0.330        | 0.288 | 0.332        |
| Electricity | <b>0.170</b> | <b>0.258</b> | 0.190        | 0.280        | <u>0.176</u> | <u>0.265</u> | 0.205    | 0.295 | 0.226        | 0.319 | 0.194    | 0.293 | 0.492      | 0.523 | 0.201        | 0.275        | 0.200    | 0.296 | 0.201        | 0.290        | 0.206 | 0.287        |
| Exchange    | <b>0.336</b> | <b>0.391</b> | 0.359        | <u>0.401</u> | 0.362        | 0.406        | 0.389    | 0.417 | <u>0.339</u> | 0.414 | 0.430    | 0.446 | 0.539      | 0.519 | 0.398        | 0.422        | 0.383    | 0.417 | 0.384        | 0.414        | 0.366 | 0.408        |
| Weather     | <b>0.233</b> | <b>0.269</b> | 0.245        | 0.274        | 0.261        | 0.281        | 0.258    | 0.279 | 0.265        | 0.316 | 0.256    | 0.283 | 0.348      | 0.382 | <u>0.242</u> | <u>0.272</u> | 0.252    | 0.277 | 0.244        | 0.274        | 0.271 | 0.291        |
| Traffic     | <b>0.407</b> | <b>0.274</b> | 0.509        | 0.307        | <u>0.422</u> | <u>0.283</u> | 0.482    | 0.309 | 0.688        | 0.428 | 0.633    | 0.334 | 1.028      | 0.605 | 0.466        | 0.287        | 0.539    | 0.364 | 0.590        | 0.373        | 0.547 | 0.345        |
| Solar       | <b>0.213</b> | <b>0.258</b> | 0.228        | 0.279        | 0.238        | <u>0.263</u> | 0.249    | 0.292 | 0.330        | 0.401 | 0.268    | 0.286 | 0.727      | 0.634 | <u>0.226</u> | 0.269        | 0.298    | 0.301 | 0.289        | 0.321        | 0.363 | 0.337        |

Table 1: Long-term forecasting results. All the results are averaged from 4 different prediction lengths {96, 192, 336, 720}, and the look-back length is fixed to 96 for all baselines. A lower MSE or MAE indicates a better prediction, with the best in boldface and second in underline.

| Models | DMSC(Ours)   |              | TimeMixer |       | iTransformer |       | PatchTST |       | Dlinear |       | TimesNet     |              | Autoformer |       | TimeXer      |              | PatchMLP |       | TimeKAN |       | AMD   |       |
|--------|--------------|--------------|-----------|-------|--------------|-------|----------|-------|---------|-------|--------------|--------------|------------|-------|--------------|--------------|----------|-------|---------|-------|-------|-------|
| Metric | MSE          | MAE          | MSE       | MAE   | MSE          | MAE   | MSE      | MAE   | MSE     | MAE   | MSE          | MAE          | MSE        | MAE   | MSE          | MAE          | MSE      | MAE   | MSE     | MAE   | MSE   | MAE   |
| PEMS03 | <b>0.131</b> | <b>0.241</b> | 0.190     | 0.288 | 0.261        | 0.328 | 0.308    | 0.368 | 0.279   | 0.377 | <u>0.154</u> | <u>0.254</u> | 0.513      | 0.517 | 0.180        | 0.280        | 0.220    | 0.289 | 0.283   | 0.357 | 0.281 | 0.368 |
| PEMS04 | <b>0.129</b> | <b>0.244</b> | 0.230     | 0.324 | 0.913        | 0.801 | 0.365    | 0.405 | 0.295   | 0.389 | <u>0.138</u> | <u>0.250</u> | 0.472      | 0.495 | 0.327        | 0.418        | 0.203    | 0.305 | 0.297   | 0.372 | 0.327 | 0.397 |
| PEMS07 | <u>0.092</u> | <u>0.193</u> | 0.165     | 0.256 | 0.109        | 0.214 | 0.438    | 0.422 | 0.329   | 0.394 | 0.109        | 0.216        | 0.418      | 0.466 | <b>0.088</b> | <b>0.192</b> | 0.166    | 0.262 | 0.264   | 0.346 | 0.196 | 0.318 |
| PEMS08 | <b>0.162</b> | <b>0.228</b> | 0.264     | 0.332 | 0.199        | 0.278 | 0.347    | 0.390 | 0.403   | 0.444 | <u>0.198</u> | <u>0.236</u> | 0.603      | 0.541 | 0.206        | 0.248        | 0.244    | 0.326 | 0.337   | 0.377 | 0.427 | 0.443 |

Table 2: Short-term forecasting results. All the results are averaged from 4 different prediction lengths {12, 24, 48, 96}, and the look-back length is fixed to 96 for all baselines.

predictions  $\hat{\mathbf{Y}}^l$  are fused using time-dependent weights:

$$\mathbf{w} = \text{Softmax}(\text{MLP}(\bigoplus_{l=1}^L \phi(\mathbf{F}_l) \cdot \mathbf{w}_{hist})), \quad (12)$$

where  $\phi(\cdot)$  denotes temporal descriptors,  $\mathbf{w}_{hist}$  denotes historical weighting memory, and  $\bigoplus$  means concatenation. The final prediction integrates multi-scale outputs:

$$\hat{\mathbf{Y}} = \sum_{l=1}^L \mathbf{w}_l \left( \sum_{m=1}^M \Omega_m^G \mathcal{G}_m(\mathbf{F}_l) + \sum_{n \in \mathcal{I}} \hat{\Omega}_n^L \mathcal{L}_n(\mathbf{F}_l) \right), \quad (13)$$

Meanwhile, to ensure balanced expert utilization, ASR-MoE incorporates a balance loss:

$$\mathcal{L}_{balance} = -\lambda \mathbf{E}[\sum_j \Omega_j \log \Omega_j], \quad (14)$$

and the overall loss function of DMSC is as follows:

$$\mathcal{L} = \mathcal{L}_{pred} + \mathcal{L}_{balance}, \quad (15)$$

where  $\mathcal{L}_{pred}$  denotes the Mean Squared Error (MSE) loss.

By combining hierarchical expert specialization, dynamic routing, and temporal-aware weighting, ASR-MoE enables the proposed framework to adaptively prioritize relevant scales and experts, achieving a synergy between long-term trend projection and short-term detail refinement in TSF.

## Experiments

To comprehensively verify the performance and effectiveness of DMSC, we conduct extensive experiments on 13 real-world TSF benchmarks spanning multiple domains and temporal resolutions.

**Datasets.** For long-term forecasting, we conduct experiments on nine well-established benchmarks, including ETT (ETTh1, ETTh2, ETTm1, ETTm2), Electricity, ECL, Traffic, Weather and Solar. For short-term forecasting, we experiment on PEMS (PEMS03, PEMS04, PEMS07, PEMS08).

**Baselines.** We compare our framework with ten SOTA models of TSF, including transformer-based models: iTransformer (Liu et al. 2024), PatchTST (Nie et al. 2023), Autoformer (Wu et al. 2021), TimeXer (Wang et al. 2024d); MLP-based models: DLinear (Zeng et al. 2023), PatchMLP (Tang and Zhang 2024), AMD (Hu et al. 2025); CNN-based models: TimesNet (Wu et al. 2023a), TimeMixer (Wang et al. 2024b), and a novel architecture TimeKAN (Huang et al. 2025).

## Main Results

The main results are listed in Table 1 for long-term forecasting and Table 2 for short-term forecasting, in which the lower MSE and MAE indicates better forecasting performance. Our proposed DMSC framework demonstrates consistent SOTA performance across all 13 benchmarks, achieving the lowest MAE and MSE in most experimental configurations. These results confirm robust effectiveness and generality of DMSC for both long-term and short-term forecasting tasks. Notably, multi-scale architectures like TimesNet and PatchMLP exhibit instability due to their inflexible fusion mechanisms that fails to fully leverage captured multiscale features. And the preferable performance of TimeXer and iTransformer underscores the effect of the

patch-wise and variate-wise strategy in information representation, but their fixed decomposition schemes inherently constrain the model’s potential. Compared to iTransformer and Dlinear, DMSC has a considerable improvement in data with more variate(Traffic, PEMS), demonstrating the necessity of jointly dependency modeling. Collectively, the above observations validate DMSC’s efficacy in addressing the core challenges of time series forecasting.

## Model Analysis

**Ablation Study.** To rigorously evaluate the contribution and effectiveness of each component within DMSC, systematic ablation experiments were conducted on Electricity, Weather, Traffic and Solar datasets. We first ablated each of the three core blocks individually. Results of Table 3 highlight the complementary synergy of the three blocks. Subsequently, we performed further ablation analyses for each block. For EMPD in Table 4, ① employs predefined static patch size for exponentially decaying. For TIB in Table 5, ② retains only intra-patch feature extraction without joint modeling of three dependency representations, while ③ removes dynamic interaction mechanism  $\mathbf{F}_{fused}^l$  in TIB. For ASR-MoE in Table 6, ④ and ⑤ eliminate global experts and local experts respectively, and ⑦ replace the prediction heads with a simple summation of multiple linear layers. Based on the above experiments, we have the following observations.

| Dataset     | Electricity  |              | Weather      |              | Traffic      |              | Solar        |              |
|-------------|--------------|--------------|--------------|--------------|--------------|--------------|--------------|--------------|
|             | MSE          | MAE          | MSE          | MAE          | MSE          | MAE          | MSE          | MAE          |
| DMSC(Ours)  | <b>0.170</b> | <b>0.258</b> | <b>0.233</b> | <b>0.269</b> | <b>0.407</b> | <b>0.274</b> | <b>0.213</b> | <b>0.258</b> |
| w/o EMPD    | 0.181        | 0.270        | 0.248        | 0.275        | 0.503        | 0.323        | 0.248        | 0.278        |
| w/o TIB     | 0.185        | 0.273        | 0.255        | 0.278        | 0.497        | 0.321        | 0.264        | 0.293        |
| w/o ASR-MoE | 0.196        | 0.285        | 0.251        | 0.280        | 0.503        | 0.320        | 0.271        | 0.305        |

Table 3: Ablation results. All results are averaged from prediction length {96, 192, 336, 720}.

| Case           | Electricity  |              | Weather      |              | Traffic      |              | Solar        |              |
|----------------|--------------|--------------|--------------|--------------|--------------|--------------|--------------|--------------|
|                | MSE          | MAE          | MSE          | MAE          | MSE          | MAE          | MSE          | MAE          |
| DMSC(Ours)     | <b>0.170</b> | <b>0.258</b> | <b>0.233</b> | <b>0.269</b> | <b>0.407</b> | <b>0.274</b> | <b>0.213</b> | <b>0.258</b> |
| ①static decomp | 0.179        | 0.271        | 0.245        | 0.272        | 0.478        | 0.313        | 0.243        | 0.276        |

Table 4: Ablation study on EMPD to compare dynamic decomposition with static predefined decomposition strategies.

| Case                        | Electricity  |              | Weather      |              | Traffic      |              | Solar        |              |
|-----------------------------|--------------|--------------|--------------|--------------|--------------|--------------|--------------|--------------|
|                             | MSE          | MAE          | MSE          | MAE          | MSE          | MAE          | MSE          | MAE          |
| DMSC(Ours)                  | <b>0.170</b> | <b>0.258</b> | <b>0.233</b> | <b>0.269</b> | <b>0.407</b> | <b>0.274</b> | <b>0.213</b> | <b>0.258</b> |
| ②only $\mathbf{F}_{intra}$  | 0.187        | 0.276        | 0.249        | 0.275        | 0.489        | 0.315        | 0.275        | 0.298        |
| ③w/o $\mathbf{F}_{fused}^l$ | 0.186        | 0.277        | 0.247        | 0.273        | 0.483        | 0.314        | 0.277        | 0.299        |

Table 5: Ablation study on TIB to validate effectiveness of joint triad dependency modeling.

| Case                 | Electricity  |              | Weather      |              | Traffic      |              | Solar        |              |
|----------------------|--------------|--------------|--------------|--------------|--------------|--------------|--------------|--------------|
|                      | MSE          | MAE          | MSE          | MAE          | MSE          | MAE          | MSE          | MAE          |
| DMSC(Ours)           | <b>0.170</b> | <b>0.258</b> | <b>0.233</b> | <b>0.269</b> | <b>0.407</b> | <b>0.274</b> | <b>0.213</b> | <b>0.258</b> |
| ④Agg. Heads          | 0.194        | 0.281        | 0.254        | 0.281        | 0.510        | 0.312        | 0.278        | 0.299        |
| ⑤w/o $\mathcal{E}^g$ | 0.191        | 0.280        | 0.250        | 0.279        | 0.504        | 0.324        | 0.265        | 0.293        |
| ⑥w/o $\mathcal{E}^l$ | 0.199        | 0.285        | 0.243        | 0.271        | 0.488        | 0.326        | 0.275        | 0.303        |

Table 6: Ablation study on ASR-MoE to assess the importance of expert specialization (global/local) and dynamic fusion strategies for prediction results.

For EMPD, employing predefined static patches or raw data embedding instead of input-adaptive decomposition degraded performance. This demonstrates the necessity of patch-wise decomposition and lightweight neural controllers for granularity adjustment. For TIB, replacing joint modeling with standard convolution layers or only intra-patch features resulted in performance drops, validating the need for comprehensive and joint modeling of triadic dependencies. Meanwhile, ablation of dynamic fusion mechanism in TIB shows its effectiveness to dynamically fuse these heterogeneous dependencies, as static handling of such relationships undermines holistic dependency capture across different scales. For ASR-MoE, substituting our adaptive prediction mechanism with either a single prediction head or summation of multiple linear heads yields substantial performance declines, which demonstrates that single heads inadequately utilize multiscale features, and simple aggregation of multiple heads overlooks the differential importance of scale-specific dependencies. Moreover, removing either global or local experts leads to performance degradation, which demonstrates the necessity of leveraging distinct experts for different temporal patterns. The elaborate design in ASR-MoE for collective utilization of specialized experts can effectively harness diverse temporal dynamics.

Collectively, these ablation studies confirm that dynamic and multiscale modeling is central to DMSC. All of three blocks and the proposed progressive cascade architecture together enable robust multiscale modeling and hierarchical feature learning. The consistent performance drops from ablating any component validate the integrity of DMSC’s architecture.

**ASR-MoE Expert Specialization.** To verify the specialization of ASR-MoE experts, we conducted activation pattern visualization across different experts. As shown in Fig. 2, distinct specialization profiles emerge: global experts exhibit pronounced sensitivity to persistent trends and long-periodic patterns, while local experts demonstrate acute responsiveness to short-periodic patterns and high-frequency fluctuations. This specialization demonstrates adaptively knowledge learning for different experts.

**Increasing Look-Back Length.** Theoretically, longer input sequences provide richer historical information, potentially enhancing the model’s ability to capture long-term trends and complex multiscale dependencies. Our DMSC framework is specifically designed to leverage this, with

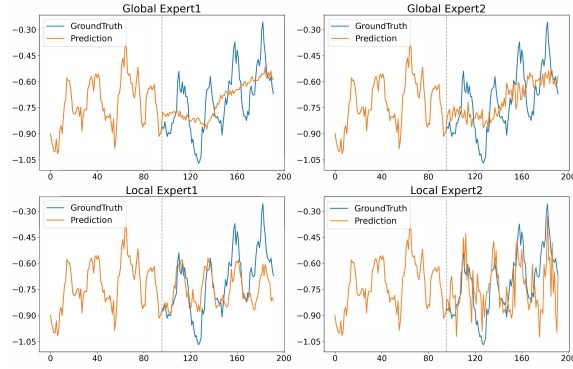


Figure 2: Visualization of different experts on ETTh1 datasets, the look-back and prediction length are set to 96.

its dynamic multiscale coordination capability inherently enabling adaptive modeling and effective utilization of information across varying look-back lengths. So we evaluate DMSC with look-back length from  $\{48, 96, 192, 336, 720\}$  on Electricity dataset. As shown in Fig. 3, DMSC consistently improves performance with longer historical sequences while outperforms other baselines, demonstrating its capacity to extract richer temporal representations from extended contexts and effectiveness in integrating multiscale temporal patterns.

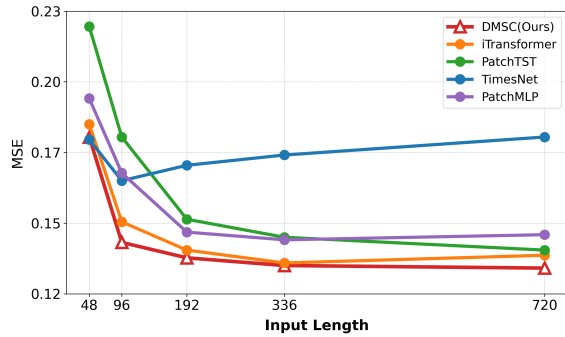


Figure 3: Forecasting results with varying look-back length on Electricity dataset. Look-back lengths are set to  $\{48, 96, 192, 336, 720\}$ , and prediction length is fixed to 96.

**Efficiency Analysis** We rigorously evaluate the memory usage and training time of DMSC against other SOTA baselines on Weather(21 variates) and Traffic datasets(862 variates). As demonstrated in Fig. 4 and Fig. 5, DMSC consistently achieves superior efficiency in both memory usage and training time while maintaining competitive forecasting accuracy. Crucially, DMSC demonstrates near-linear scalability to long sequences—significantly outperforming Transformer-based models that exhibit quadratic complexity and MLP-based approaches with parameter explosion. These efficiency gains are particularly pronounced when handling high-dimensional multivariate data (Traffic), where DMSC’s cross-variable interaction avoids the computational

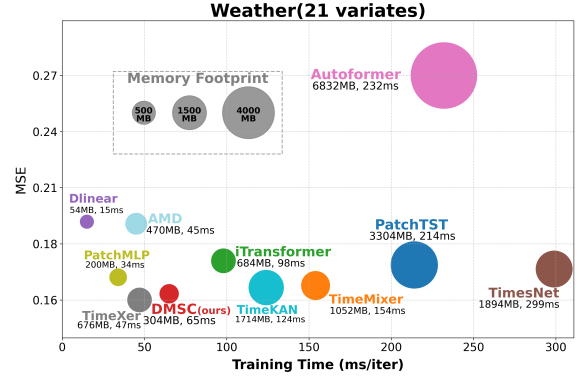


Figure 4: Model efficiency analysis under 96-look-back length and 96-prediction length on Weather(21 variates) datasets. Batch size is set to 128.

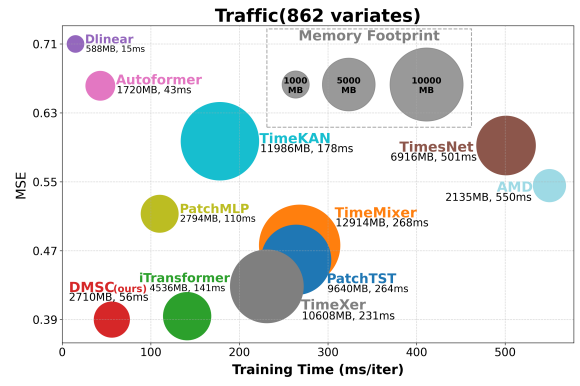


Figure 5: Model efficiency analysis under 96-look-back length and 96-prediction length on Traffic(862 variates) datasets. Batch size is set to 16.

overhead of exhaustive pairwise attention while maintaining modeling capacity. Our framework thus establishes a balance of accuracy, latency, and memory usage for practical deployment scenarios.

## Conclusion

This paper introduces the Dynamic Multi-Scale Interaction (DMSC) framework, a novel framework that can advance TSF through dynamic multiscale coordination. DMSC has three key contributions: 1) an Embedded Multi-Scale Patch Decomposition(EMPD) is introduced to dynamically decompose time series into hierarchical patches to eliminate predefined and fixed scale constraints; 2) Triad Interaction Block (TIB) is proposed to jointly model intra-patch, inter-patch, and cross-variable dependencies, thus forming a coarse-to-fine feature pyramid through progressive cascade layers; 3) an Adaptive Scale Routing MoE (ASR-MoE) is proposed to dynamically fuse multiscale predictions via temporal-aware weighting of global and local experts. Extensive experiments on thirteen benchmarks demonstrate that DMSC can achieve SOTA performance and superior efficiency. Future work will extend the framework to multi-



tasks learning and explore optimizations on intricate dataset environments, improving its multi-scenario applicability for real world.

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## Implementation Details

### Dataset Descriptions

We conduct long-term forecasting experiments on 9 real-world datasets, including ETT(ETTh1, ETTh2, ETTm1, ETTm2); Electricity(ECL); Exchange; Solar; Traffic; and

Weather datasets. Meanwhile, we use PEMS(PEMS03, PEMS04, PEMS07, PEMS08) datasets to conduct short-term forecasting experiments. The details of datasets are listed in Tab 7.

### Metrics Details

To evaluate model’s performance for TSF, we utilize the mean square error(MSE) and mean absolute error(MAE). The calculations of metrics are as follows:

$$\text{MSE} = \frac{1}{n} \sum_{i=1}^n (\mathbf{Y}_i - \hat{\mathbf{Y}}_i)^2 \quad (16)$$

$$\text{MAE} = \frac{1}{n} \sum_{i=1}^n |\mathbf{Y}_i - \hat{\mathbf{Y}}_i| \quad (17)$$

### Implementation Details

All experiments are implemented in Pytorch, and conducted on a single NVIDIA RTX 4090 24GB GPU. We utilize ADAM optimizer with an initial learning rate  $10^{-3}$  and L2 loss for model optimization. For DMSC, we set layers of progressive cascade framework to 2 - 5, embedding dimension is set to 64, 128, 256, 512. The patch decay rate is set to 2 for exponentially degradation. All experiments are based on the framework of TimesNet. And all the baselines are implemented based on the configurations of original paper and its code.

## Model Analysis

### Model Complexity Analysis

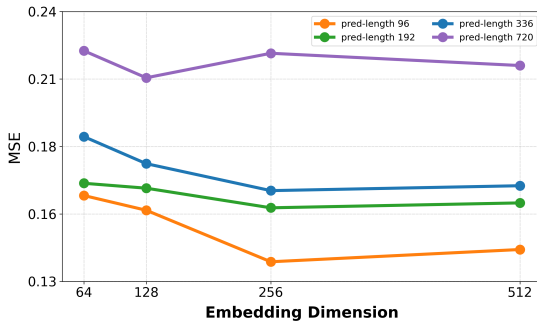
The computational complexity of DMSC is dominated by three core components: EMPD, TIB, and ASR-MoE. EMPD hierarchically decomposes input sequences into exponentially scaled patches via unfolding and linear projection, yielding a lightweight  $O(C \cdot L \cdot d_{\text{model}})$  complexity, where  $C$  is the number of variate,  $L$  is the sequence length, and  $d_{\text{model}}$  is the embedding dimension. TIB employs depth-wise separable convolutions and dilated convolutions to model intra-patch, inter-patch, and cross-variable dependencies, maintaining near-linear complexity  $O(C \cdot N \cdot d_{\text{model}}^2)$  per layer ( $N$  is patch count). ASR-MoE leverages sparse activation, routing each input to only top- $K$  experts (global and local), reducing fusion complexity from quadratic to  $O(B \cdot C \cdot d_{\text{model}} \cdot (K + S))$  ( $B$ : batch size,  $S$ : shared experts). Collectively, DMSC achieves  $O(L)$  sequential scalability, outperforming Transformer-based models( $O(L^2)$ ) and MLP-based alternatives( $O(L \cdot d_{ff}^2)$ ), while excelling in both long- and short-term forecasting tasks.

### Hyperparameter Sensitivity Analysis

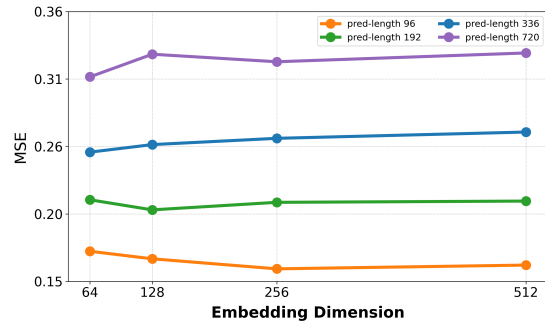
**Embedding Dimension.** The embedding dimension determines the richness of feature representations. We set  $d_{\text{model}} \in \{64, 128, 256, 512\}$  to evaluate model’s performance on ECL and Weather dataset. The results are shown in Fig. 6a and Fig. 6b, we choose the optimal dimension at 128 / 256 - smaller embedding impairs multiscale feature separation, while larger configuration yield marginal gains with higher memory cost, indicating dimensionality saturating.

| Name         | Domain         | Length | Num | Prediction Length | Dataset Size        | Freq. (m) |
|--------------|----------------|--------|-----|-------------------|---------------------|-----------|
| ETTh1        | Temperature    | 14400  | 7   | {96,192,336,720}  | (8545,2881,2881)    | 60        |
| ETTh2        | Temperature    | 14400  | 7   | {96,192,336,720}  | (8545,2881,2881)    | 60        |
| ETTm1        | Temperature    | 57600  | 7   | {96,192,336,720}  | (34465,11521,11521) | 15        |
| ETTm2        | Temperature    | 57600  | 7   | {96,192,336,720}  | (34465,11521,11521) | 15        |
| Electricity  | Electricity    | 26304  | 321 | {96,192,336,720}  | (18317,2633,5261)   | 60        |
| Exchange     | Exchange Rate  | 7588   | 8   | {96,192,336,720}  | (5120,665,1422)     | 1440      |
| Traffic      | Road Occupancy | 17544  | 862 | {96,192,336,720}  | (12185,1757,3509)   | 60        |
| Weather      | Weather        | 52696  | 21  | {96,192,336,720}  | (36792,5271,10540)  | 10        |
| Solar-Energy | Energy         | 52179  | 137 | {96,192,336,720}  | (36601,5161,10417)  | 10        |
| PEMS03       | Traffic Flow   | 26208  | 358 | {12,24,48,96}     | (15617,5135,5135)   | 5         |
| PEMS04       | Traffic Flow   | 16992  | 307 | {12,24,48,96}     | (10172,3375,3375)   | 5         |
| PEMS07       | Traffic Flow   | 28224  | 883 | {12,24,48,96}     | (16711,5622,5622)   | 5         |
| PEMS08       | Traffic Flow   | 17856  | 170 | {12,24,48,96}     | (10690,3548,3548)   | 5         |

Table 7: Dataset Descriptions. Num is the number of variable. Dataset size is organized in (Train, Validation, Test).

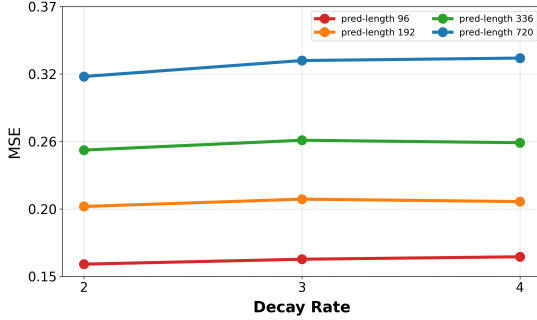


(a) Performance on Electricity dataset.

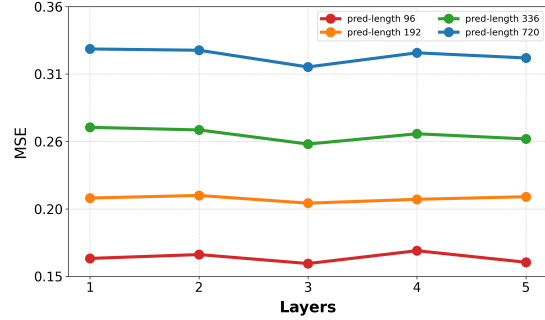


(b) Performance on Weather dataset.

Figure 6: Hyperparameter sensitivity analysis under different embedding dimension,  $d_{\text{model}}$  is set to  $\{64, 128, 256, 512\}$ . Left is conducted on Electricity dataset and right is on Weather dataset. Look-back length is fixed to 96, and prediction lengths are set to  $\{96, 192, 336, 720\}$ .



(a) Performance of different patch length decay rate.



(b) Performance of different progressive cascade layers.

Figure 7: Hyperparameter sensitivity analysis on Weather dataset. Left shows performance of different patch length decay rate, right shows performance of different progressive cascade layers. Look-back length is fixed to 96, and prediction lengths are set to  $\{96, 192, 336, 720\}$ .

**EMPD Patch Length Decay.** EMPD employs exponentially decaying patch lengths to capture multiscale dependencies. We vary the decay rate (2, 3, 4) and find that a decay factor of 2 yields the best results from Fig. 7a on weather dataset. It can be seen that a decay factor of 2 yields the best

forecasting performance, which balances the preservation of sufficient local details in fine-grained layers against the extraction of broader trends in coarse-grained layers.

**Number of Progressive Cascade Layers.** The number of cascade layers ( $l \in \{1 - 5\}$ ) affects the hierarchical feature extraction. As shown in Fig. 7b, three layers can achieve peak efficiency - shallower stacks (1 - 2 layers) preserve perfect accuracy via residual propagation but miss fine-grained interactions, while deeper configurations (4 - 5 layers) introduce diminishing returns with much more latency but fewer improvement.

## Full Results

We evaluate DMSC on 13 real-world TSF benchmarks spanning diverse domains. Tab 8 shows the full results of long-term forecasting tasks on ETT (ETTh1, ETTh2, ETTm1, ETTm2), Electricity, ECL, Traffic, Weather and Solar datasets. Tab 9 contains the full results of short-term forecasting tasks on PEMS(PEMS03, PEMS04, PEMS07, PEMS08) datasets. From full results, we can see that DMSC achieves pretty good performance on most datasets and tasks.

Meanwhile, Tab 10 shows the full results of ablation study on Electricity, Weather, Traffic and Solar datasets, which demonstrating the efficiency and contribution of each component in DMSC.

| Models      |     | DMSC(Ours)   |              | TimeMixer    |              | iTransformer |              | PatchTST |       | Dlinear      |       | TimesNet |       | Autoformer |       | TimeXer      |              | PatchMLP |              | TimeKAN      |              | AMD   |              |
|-------------|-----|--------------|--------------|--------------|--------------|--------------|--------------|----------|-------|--------------|-------|----------|-------|------------|-------|--------------|--------------|----------|--------------|--------------|--------------|-------|--------------|
| Metric      |     | MSE          | MAE          | MSE          | MAE          | MSE          | MAE          | MSE      | MAE   | MSE          | MAE   | MSE      | MAE   | MSE        | MAE   | MSE          | MAE          | MSE      | MAE          | MSE          | MAE          | MSE   | MAE          |
| ETTh1       | 96  | <b>0.370</b> | <b>0.395</b> | 0.379        | 0.397        | 0.387        | 0.405        | 0.378    | 0.399 | 0.397        | 0.412 | 0.415    | 0.429 | 0.589      | 0.526 | 0.386        | 0.404        | 0.393    | 0.405        | 0.387        | 0.401        | 0.394 | 0.404        |
|             | 192 | <b>0.408</b> | <b>0.420</b> | 0.430        | 0.429        | 0.441        | 0.436        | 0.427    | 0.429 | 0.446        | 0.441 | 0.479    | 0.466 | 0.653      | 0.551 | 0.429        | 0.435        | 0.443    | 0.434        | <u>0.415</u> | <u>0.423</u> | 0.444 | 0.432        |
|             | 336 | <b>0.421</b> | <b>0.433</b> | 0.493        | 0.459        | 0.494        | 0.463        | 0.468    | 0.455 | 0.489        | 0.467 | 0.517    | 0.482 | 0.715      | 0.581 | 0.484        | 0.457        | 0.486    | 0.456        | <u>0.453</u> | <u>0.443</u> | 0.485 | 0.451        |
|             | 720 | <u>0.462</u> | <b>0.457</b> | 0.522        | 0.493        | 0.488        | 0.483        | 0.508    | 0.497 | 0.513        | 0.511 | 0.505    | 0.490 | 0.726      | 0.601 | 0.544        | 0.513        | 0.509    | 0.487        | <b>0.461</b> | <u>0.463</u> | 0.486 | 0.472        |
|             | Avg | <b>0.415</b> | <b>0.426</b> | 0.456        | 0.444        | 0.452        | 0.446        | 0.445    | 0.445 | 0.461        | 0.457 | 0.479    | 0.466 | 0.670      | 0.564 | 0.460        | 0.452        | 0.458    | 0.445        | <u>0.429</u> | <u>0.432</u> | 0.452 | 0.439        |
| ETT2        | 96  | <b>0.275</b> | <b>0.329</b> | 0.290        | 0.341        | 0.301        | 0.350        | 0.295    | 0.347 | 0.341        | 0.394 | 0.316    | 0.358 | 0.443      | 0.459 | <u>0.284</u> | <u>0.337</u> | 0.311    | 0.358        | 0.291        | 0.342        | 0.397 | 0.451        |
|             | 192 | <b>0.359</b> | <b>0.383</b> | 0.366        | 0.394        | 0.380        | 0.399        | 0.378    | 0.401 | 0.482        | 0.479 | 0.415    | 0.414 | 0.500      | 0.506 | <u>0.366</u> | <u>0.391</u> | 0.404    | 0.415        | 0.376        | 0.393        | 0.501 | 0.501        |
|             | 336 | <b>0.376</b> | <b>0.398</b> | 0.425        | 0.433        | <u>0.423</u> | <u>0.431</u> | 0.425    | 0.442 | 0.591        | 0.541 | 0.452    | 0.448 | 0.506      | 0.502 | 0.438        | 0.438        | 0.447    | 0.451        | 0.437        | 0.443        | 0.611 | 0.563        |
|             | 720 | 0.412        | <b>0.424</b> | <b>0.405</b> | <u>0.431</u> | 0.431        | 0.447        | 0.436    | 0.456 | 0.839        | 0.661 | 0.461    | 0.463 | 0.503      | 0.509 | <u>0.407</u> | 0.449        | 0.462    | 0.464        | 0.451        | 0.458        | 0.956 | 0.718        |
|             | Avg | <b>0.355</b> | <b>0.383</b> | <u>0.372</u> | <u>0.400</u> | 0.384        | 0.407        | 0.383    | 0.412 | 0.563        | 0.519 | 0.411    | 0.421 | 0.488      | 0.494 | 0.374        | 0.404        | 0.406    | 0.422        | 0.389        | 0.409        | 0.616 | 0.558        |
| ETTm1       | 96  | <b>0.304</b> | <b>0.335</b> | 0.319        | 0.359        | 0.341        | 0.376        | 0.326    | 0.366 | 0.346        | 0.374 | 0.336    | 0.375 | 0.564      | 0.506 | <u>0.318</u> | <u>0.356</u> | 0.321    | 0.362        | 0.326        | 0.365        | 0.331 | 0.363        |
|             | 192 | <b>0.346</b> | <b>0.364</b> | <u>0.360</u> | 0.384        | 0.382        | 0.396        | 0.365    | 0.387 | 0.382        | 0.391 | 0.377    | 0.395 | 0.586      | 0.516 | 0.373        | 0.389        | 0.364    | <u>0.381</u> | <u>0.360</u> | 0.384        | 0.373 | 0.382        |
|             | 336 | <u>0.372</u> | <u>0.392</u> | 0.395        | 0.406        | 0.420        | 0.421        | 0.392    | 0.406 | 0.415        | 0.415 | 0.418    | 0.420 | 0.679      | 0.547 | 0.412        | <b>0.387</b> | 0.396    | 0.404        | <b>0.369</b> | 0.402        | 0.405 | 0.403        |
|             | 720 | <u>0.451</u> | <u>0.440</u> | 0.461        | 0.444        | 0.487        | 0.456        | 0.461    | 0.443 | 0.473        | 0.451 | 0.541    | 0.481 | 0.715      | 0.567 | 0.460        | 0.450        | 0.468    | 0.443        | <b>0.449</b> | <b>0.437</b> | 0.467 | 0.437        |
|             | Avg | <b>0.368</b> | <b>0.383</b> | 0.384        | 0.398        | 0.408        | 0.412        | 0.386    | 0.400 | 0.404        | 0.408 | 0.418    | 0.418 | 0.636      | 0.534 | 0.391        | 0.395        | 0.387    | 0.398        | <u>0.376</u> | 0.397        | 0.394 | <u>0.396</u> |
| ETTm2       | 96  | <u>0.174</u> | <u>0.256</u> | 0.179        | 0.261        | 0.186        | 0.272        | 0.184    | 0.269 | 0.193        | 0.293 | 0.188    | 0.268 | 0.554      | 0.469 | <b>0.172</b> | <b>0.254</b> | 0.176    | 0.259        | 0.176        | 0.263        | 0.187 | 0.271        |
|             | 192 | <b>0.233</b> | <b>0.295</b> | <u>0.238</u> | <u>0.301</u> | 0.252        | 0.312        | 0.247    | 0.307 | 0.284        | 0.361 | 0.250    | 0.306 | 0.609      | 0.497 | 0.241        | 0.302        | 0.246    | 0.304        | 0.239        | <u>0.301</u> | 0.251 | 0.309        |
|             | 336 | <b>0.296</b> | <b>0.331</b> | <u>0.299</u> | <u>0.339</u> | 0.315        | 0.351        | 0.313    | 0.354 | 0.382        | 0.429 | 0.306    | 0.341 | 0.401      | 0.409 | 0.301        | 0.340        | 0.309    | 0.344        | 0.304        | 0.346        | 0.309 | 0.346        |
|             | 720 | <b>0.370</b> | <b>0.385</b> | <u>0.395</u> | <u>0.394</u> | 0.415        | 0.408        | 0.409    | 0.406 | 0.558        | 0.525 | 0.420    | 0.405 | 0.443      | 0.433 | <u>0.394</u> | 0.395        | 0.417    | 0.408        | 0.410        | 0.408        | 0.407 | 0.400        |
|             | Avg | <b>0.268</b> | <b>0.317</b> | 0.278        | 0.324        | 0.292        | 0.336        | 0.288    | 0.334 | 0.354        | 0.402 | 0.291    | 0.330 | 0.502      | 0.452 | <u>0.277</u> | <u>0.323</u> | 0.287    | 0.329        | 0.282        | 0.330        | 0.288 | 0.332        |
| Electricity | 96  | <b>0.138</b> | <b>0.223</b> | 0.161        | 0.252        | <u>0.148</u> | <u>0.241</u> | 0.181    | 0.274 | 0.211        | 0.302 | 0.163    | 0.267 | 0.232      | 0.347 | 0.241        | 0.244        | 0.167    | 0.264        | 0.177        | 0.270        | 0.185 | 0.267        |
|             | 192 | <u>0.160</u> | <b>0.258</b> | 0.176        | 0.269        | 0.167        | 0.248        | 0.187    | 0.280 | 0.211        | 0.305 | 0.184    | 0.284 | 0.363      | 0.447 | <b>0.159</b> | <u>0.260</u> | 0.181    | 0.276        | 0.185        | 0.276        | 0.190 | 0.272        |
|             | 336 | <b>0.167</b> | <b>0.253</b> | 0.193        | 0.283        | 0.179        | <u>0.271</u> | 0.204    | 0.296 | 0.223        | 0.319 | 0.196    | 0.297 | 0.599      | 0.595 | <u>0.177</u> | 0.276        | 0.203    | 0.303        | 0.201        | 0.292        | 0.205 | 0.288        |
|             | 720 | <u>0.213</u> | <u>0.299</u> | 0.232        | 0.316        | <b>0.208</b> | <b>0.298</b> | 0.246    | 0.328 | 0.258        | 0.351 | 0.232    | 0.325 | 0.775      | 0.701 | 0.229        | 0.321        | 0.251    | 0.341        | 0.241        | 0.323        | 0.246 | 0.321        |
|             | Avg | <b>0.170</b> | <b>0.258</b> | 0.190        | 0.280        | <u>0.176</u> | <u>0.265</u> | 0.205    | 0.295 | 0.226        | 0.319 | 0.194    | 0.293 | 0.492      | 0.523 | 0.201        | 0.275        | 0.200    | 0.296        | 0.201        | 0.290        | 0.206 | 0.287        |
| Exchange    | 96  | <b>0.082</b> | <b>0.201</b> | <u>0.087</u> | <u>0.204</u> | 0.088        | 0.208        | 0.094    | 0.213 | 0.098        | 0.233 | 0.115    | 0.242 | 0.158      | 0.290 | 0.094        | 0.214        | 0.094    | 0.217        | 0.092        | 0.212        | 0.088 | 0.208        |
|             | 192 | <b>0.174</b> | <u>0.298</u> | <u>0.177</u> | <b>0.295</b> | 0.180        | 0.303        | 0.182    | 0.303 | 0.186        | 0.325 | 0.216    | 0.333 | 0.299      | 0.406 | 0.182        | 0.303        | 0.187    | 0.311        | 0.180        | 0.300        | 0.182 | 0.305        |
|             | 336 | <b>0.315</b> | <b>0.405</b> | 0.328        | <u>0.414</u> | 0.331        | 0.418        | 0.347    | 0.426 | <u>0.325</u> | 0.434 | 0.375    | 0.444 | 0.470      | 0.511 | 0.384        | 0.448        | 0.342    | 0.424        | 0.352        | 0.430        | 0.332 | 0.417        |
|             | 720 | <u>0.772</u> | <b>0.659</b> | 0.847        | 0.692        | 0.848        | 0.695        | 0.931    | 0.724 | <b>0.746</b> | 0.663 | 1.012    | 0.765 | 1.228      | 0.869 | 0.932        | 0.724        | 0.908    | 0.715        | 0.912        | 0.715        | 0.861 | <u>0.701</u> |
|             | Avg | <b>0.336</b> | <b>0.391</b> | 0.359        | <u>0.401</u> | 0.362        | 0.406        | 0.389    | 0.417 | <u>0.339</u> | 0.414 | 0.430    | 0.446 | 0.539      | 0.519 | 0.398        | 0.422        | 0.383    | 0.417        | 0.384        | 0.414        | 0.366 | 0.408        |
| Weather     | 96  | <u>0.160</u> | <u>0.210</u> | 0.164        | <u>0.210</u> | 0.176        | 0.216        | 0.174    | 0.215 | 0.195        | 0.252 | 0.172    | 0.221 | 0.301      | 0.364 | <b>0.157</b> | <b>0.205</b> | 0.168    | 0.214        | 0.163        | 0.210        | 0.194 | 0.236        |
|             | 192 | <u>0.207</u> | <u>0.250</u> | 0.208        | 0.251        | 0.225        | 0.257        | 0.221    | 0.256 | 0.239        | 0.299 | 0.220    | 0.260 | 0.335      | 0.385 | <b>0.204</b> | <b>0.248</b> | 0.215    | 0.255        | 0.209        | 0.251        | 0.239 | 0.271        |
|             | 336 | <b>0.253</b> | <b>0.284</b> | 0.264        | 0.292        | 0.281        | 0.299        | 0.280    | 0.297 | 0.282        | 0.333 | 0.280    | 0.302 | 0.352      | 0.376 | 0.264        | 0.293        | 0.272    | 0.295        | <u>0.263</u> | <u>0.291</u> | 0.290 | 0.306        |
|             | 720 | <b>0.313</b> | <b>0.332</b> | 0.343        | <u>0.343</u> | 0.361        | 0.353        | 0.357    | 0.349 | 0.345        | 0.381 | 0.353    | 0.350 | 0.405      | 0.401 | 0.343        | <u>0.343</u> | 0.351    | 0.346        | <u>0.341</u> | 0.344        | 0.362 | 0.352        |
|             | Avg | <b>0.233</b> | <b>0.269</b> | 0.245        | 0.274        | 0.261        | 0.281        | 0.258    | 0.279 | 0.265        | 0.316 | 0.256    | 0.283 | 0.348      | 0.382 | <u>0.242</u> | <u>0.272</u> | 0.252    | 0.277        | 0.244        | 0.274        | 0.271 | 0.291        |
| Traffic     | 96  | <b>0.389</b> | <b>0.259</b> | 0.476        | 0.292        | 0.393        | <u>0.269</u> | 0.459    | 0.299 | 0.712        | 0.438 | 0.593    | 0.317 | 0.663      | 0.403 | 0.428        | 0.271        | 0.513    | 0.352        | 0.598        | 0.382        | 0.546 | 0.346        |
|             | 192 | <b>0.405</b> | <b>0.258</b> | 0.501        | 0.301        | <u>0.412</u> | <u>0.277</u> | 0.469    | 0.303 | 0.662        | 0.417 | 0.618    | 0.327 | 0.915      | 0.557 | 0.447        | 0.280        | 0.509    | 0.350        | 0.579        | 0.365        | 0.529 | 0.335        |
|             | 336 | <b>0.398</b> | <u>0.286</u> | 0.514        | 0.314        | <u>0.424</u> | <b>0.283</b> | 0.483    | 0.309 | 0.669        | 0.419 | 0.642    | 0.341 | 1.217      | 0.704 | 0.472        | 0.289        | 0.533    | 0.360        | 0.572        | 0.361        | 0.540 | 0.339        |
|             | 720 | <b>0.437</b> | <b>0.290</b> | 0.545        | 0.320        | <u>0.459</u> | <u>0.301</u> | 0.518    | 0.326 | 0.709        | 0.437 | 0.679    | 0.350 | 1.317      | 0.755 | 0.517        | 0.307        | 0.599    | 0.395        | 0.609        | 0.384        | 0.576 | 0.358        |
|             | Avg | <b>0.407</b> | <b>0.274</b> | 0.509        | 0.307        | <u>0.422</u> | <u>0.283</u> | 0.482    | 0.309 | 0.688        | 0.428 | 0.633    | 0.334 | 1.028      | 0.605 | 0.466        | 0.287        | 0.539    | 0.364        | 0.590        | 0.373        | 0.547 | 0.345        |
| Solar       | 96  | <b>0.187</b> | <b>0.231</b> | <u>0.196</u> | 0.263        | 0.207        | <u>0.237</u> | 0.216    | 0.274 | 0.290        | 0.378 | 0.223    | 0.256 | 0.552      | 0.524 | 0.198        | 0.244        | 0.239    | 0.272        | 0.248        | 0.302        | 0.310 | 0.312        |
|             | 192 | <b>0.216</b> | <b>0.261</b> | 0.245        | 0.279        | 0.242        | <u>0.264</u> | 0.250    | 0.294 | 0.320        | 0.398 | 0.262    | 0.272 | 0.696      | 0.605 | <u>0.226</u> | 0.270        | 0.291    | 0.298        | 0.291        | 0.321        | 0.348 | 0.330        |
|             | 336 | <b>0.223</b> | <b>0.269</b> | <u>0.235</u> | 0.287        | 0.251        | <u>0.274</u> | 0.265    | 0.302 | 0.353        | 0.415 | 0.287    | 0.299 | 0.816      | 0.677 | 0.239        | 0.281        | 0.325    | 0.316        | 0.307        | 0.332        | 0.399 | 0.355        |
|             | 720 | <b>0.226</b> | <b>0.271</b> | <u>0.236</u> | 0.286        | 0.251        | 0.276        | 0.266    | 0.298 | 0.357        | 0.413 | 0.298    | 0.318 | 0.844      | 0.731 | 0.242        | <u>0.282</u> | 0.336    | 0.321        | 0.310        | 0.329        | 0.393 | 0.349        |
|             | Avg | <b>0.213</b> | <b>0.258</b> | 0.228        | 0.279        | 0.238        | <u>0.263</u> | 0.249    | 0.292 | 0.330        | 0.401 | 0.268    | 0.286 | 0.727      | 0.634 | <u>0.226</u> | 0.269        | 0.298    | 0.301        | 0.289        | 0.321        | 0.363 | 0.337        |

Table 8: Full results of long-term forecasting. All the results are selected from 4 different prediction lengths {96, 192, 336, 720}, and the look-back length is fixed to 96 for all baselines. A lower MSE or MAE indicates a better prediction, with the best in **boldface** and second in underline.

| Models | DMSC(Ours) |              | TimeMixer    |       | iTransformer |              | PatchTST     |       | Dlinear |       | TimesNet |              | Autoformer   |       | TimeXer |              | PatchMLP     |       | TimeKAN |       | AMD   |       |       |
|--------|------------|--------------|--------------|-------|--------------|--------------|--------------|-------|---------|-------|----------|--------------|--------------|-------|---------|--------------|--------------|-------|---------|-------|-------|-------|-------|
| Metric | MSE        | MAE          | MSE          | MAE   | MSE          | MAE          | MSE          | MAE   | MSE     | MAE   | MSE      | MAE          | MSE          | MAE   | MSE     | MAE          | MSE          | MAE   | MSE     | MAE   | MSE   | MAE   |       |
| PEMS03 | 12         | <b>0.066</b> | <b>0.171</b> | 0.084 | 0.194        | 0.069        | <u>0.175</u> | 0.105 | 0.216   | 0.122 | 0.245    | 0.088        | 0.195        | 0.224 | 0.346   | <u>0.068</u> | 0.179        | 0.118 | 0.191   | 0.095 | 0.210 | 0.106 | 0.225 |
|        | 24         | <u>0.092</u> | <b>0.203</b> | 0.130 | 0.244        | 0.099        | 0.210        | 0.198 | 0.296   | 0.202 | 0.320    | 0.118        | 0.224        | 0.492 | 0.513   | <b>0.089</b> | <u>0.204</u> | 0.204 | 0.231   | 0.166 | 0.281 | 0.174 | 0.294 |
|        | 48         | <b>0.136</b> | <b>0.251</b> | 0.218 | 0.317        | 0.164        | 0.275        | 0.472 | 0.466   | 0.334 | 0.428    | 0.169        | 0.268        | 0.392 | 0.459   | <u>0.137</u> | <u>0.253</u> | 0.213 | 0.314   | 0.314 | 0.393 | 0.333 | 0.417 |
|        | 96         | <b>0.230</b> | <u>0.339</u> | 0.327 | 0.398        | 0.711        | 0.651        | 0.458 | 0.493   | 0.459 | 0.517    | <u>0.239</u> | <b>0.330</b> | 0.944 | 0.749   | 0.427        | 0.483        | 0.347 | 0.421   | 0.558 | 0.543 | 0.510 | 0.536 |
|        | Avg        | <b>0.131</b> | <b>0.241</b> | 0.190 | 0.288        | 0.261        | 0.328        | 0.308 | 0.368   | 0.279 | 0.377    | <u>0.154</u> | <u>0.254</u> | 0.513 | 0.517   | 0.180        | 0.280        | 0.220 | 0.289   | 0.283 | 0.357 | 0.281 | 0.368 |
| PEMS04 | 12         | <b>0.078</b> | <b>0.183</b> | 0.105 | 0.216        | 0.766        | 0.709        | 0.116 | 0.230   | 0.147 | 0.272    | <u>0.092</u> | <u>0.202</u> | 0.211 | 0.341   | 0.293        | 0.397        | 0.109 | 0.204   | 0.107 | 0.222 | 0.131 | 0.256 |
|        | 24         | <b>0.102</b> | <b>0.215</b> | 0.168 | 0.280        | 0.799        | 0.728        | 0.216 | 0.314   | 0.225 | 0.340    | <u>0.111</u> | <u>0.224</u> | 0.394 | 0.471   | 0.308        | 0.409        | 0.129 | 0.248   | 0.178 | 0.294 | 0.196 | 0.311 |
|        | 48         | <b>0.147</b> | <b>0.261</b> | 0.270 | 0.359        | 1.041        | 0.882        | 0.503 | 0.489   | 0.356 | 0.437    | <u>0.152</u> | <u>0.266</u> | 0.429 | 0.463   | 0.339        | 0.425        | 0.213 | 0.326   | 0.329 | 0.409 | 0.344 | 0.420 |
|        | 96         | <b>0.190</b> | <u>0.316</u> | 0.377 | 0.439        | 1.045        | 0.886        | 0.623 | 0.586   | 0.453 | 0.505    | <u>0.197</u> | <b>0.308</b> | 0.853 | 0.703   | 0.367        | 0.441        | 0.361 | 0.441   | 0.572 | 0.563 | 0.638 | 0.602 |
|        | Avg        | <b>0.129</b> | <b>0.244</b> | 0.230 | 0.324        | 0.913        | 0.801        | 0.365 | 0.405   | 0.295 | 0.389    | <u>0.138</u> | <u>0.250</u> | 0.472 | 0.495   | 0.327        | 0.418        | 0.203 | 0.305   | 0.297 | 0.372 | 0.327 | 0.397 |
| PEMS07 | 12         | <b>0.059</b> | <b>0.157</b> | 0.070 | 0.173        | 0.068        | 0.169        | 0.093 | 0.206   | 0.116 | 0.241    | 0.075        | 0.179        | 0.207 | 0.335   | <u>0.061</u> | <u>0.165</u> | 0.107 | 0.178   | 0.085 | 0.199 | 0.096 | 0.223 |
|        | 24         | <u>0.078</u> | <u>0.179</u> | 0.109 | 0.215        | 0.087        | 0.190        | 0.195 | 0.295   | 0.209 | 0.327    | 0.083        | 0.198        | 0.314 | 0.412   | <b>0.071</b> | <b>0.177</b> | 0.112 | 0.216   | 0.149 | 0.268 | 0.231 | 0.360 |
|        | 48         | <u>0.104</u> | <u>0.211</u> | 0.199 | 0.296        | 0.122        | 0.231        | 0.485 | 0.469   | 0.397 | 0.456    | 0.128        | 0.235        | 0.595 | 0.553   | <b>0.100</b> | <b>0.208</b> | 0.168 | 0.282   | 0.292 | 0.383 | 0.143 | 0.272 |
|        | 96         | <u>0.128</u> | <u>0.226</u> | 0.283 | 0.341        | 0.159        | 0.267        | 0.979 | 0.716   | 0.592 | 0.552    | 0.150        | 0.253        | 0.556 | 0.563   | <b>0.120</b> | <b>0.221</b> | 0.275 | 0.373   | 0.531 | 0.535 | 0.312 | 0.417 |
|        | Avg        | <u>0.092</u> | <u>0.193</u> | 0.165 | 0.256        | 0.109        | 0.214        | 0.438 | 0.422   | 0.329 | 0.394    | 0.109        | 0.216        | 0.418 | 0.466   | <b>0.088</b> | <b>0.192</b> | 0.166 | 0.262   | 0.264 | 0.346 | 0.196 | 0.318 |
| PEMS08 | 12         | <b>0.076</b> | <b>0.178</b> | 0.100 | 0.208        | <u>0.081</u> | <u>0.183</u> | 0.109 | 0.222   | 0.153 | 0.259    | 0.158        | 0.192        | 0.295 | 0.391   | 0.146        | 0.198        | 0.096 | 0.206   | 0.105 | 0.219 | 0.141 | 0.262 |
|        | 24         | 0.124        | 0.223        | 0.171 | 0.279        | <u>0.118</u> | <u>0.222</u> | 0.205 | 0.306   | 0.238 | 0.357    | <b>0.112</b> | <b>0.219</b> | 0.345 | 0.419   | 0.171        | 0.221        | 0.144 | 0.256   | 0.179 | 0.291 | 0.243 | 0.350 |
|        | 48         | <b>0.172</b> | <u>0.216</u> | 0.317 | 0.380        | <u>0.202</u> | 0.292        | 0.493 | 0.484   | 0.473 | 0.515    | 0.231        | <b>0.198</b> | 0.503 | 0.495   | 0.220        | 0.270        | 0.253 | 0.351   | 0.343 | 0.405 | 0.439 | 0.476 |
|        | 96         | <b>0.278</b> | <b>0.296</b> | 0.468 | 0.459        | 0.395        | 0.415        | 0.582 | 0.548   | 0.748 | 0.646    | <u>0.291</u> | 0.338        | 1.268 | 0.857   | 0.285        | <u>0.303</u> | 0.482 | 0.492   | 0.720 | 0.592 | 0.886 | 0.684 |
|        | Avg        | <b>0.162</b> | <b>0.228</b> | 0.264 | 0.332        | 0.199        | 0.278        | 0.347 | 0.390   | 0.403 | 0.444    | <u>0.198</u> | <u>0.236</u> | 0.603 | 0.541   | 0.206        | 0.248        | 0.244 | 0.326   | 0.337 | 0.377 | 0.427 | 0.443 |

Table 9: Full results of short-term forecasting. All the results are selected from 4 different prediction lengths {12, 24, 48, 96}, and the look-back length is fixed to 96 for all baselines. A lower MSE or MAE indicates a better prediction, with the best in **boldface** and second in underline.

| Design      | DMSC(Ours) |              | w/o EMPD     |       | w/o TIB |       | w/o ASR-MoE |       | ①static decomp |       | ②only $\mathbf{F}_{intra}$ |       | ③w/o $\mathbf{F}_{fused}^l$ |       | ④Agg. Heads |       | ⑤w/o $\mathcal{E}^g$ |       | ⑥w/o $\mathcal{E}^l$ |       |       |
|-------------|------------|--------------|--------------|-------|---------|-------|-------------|-------|----------------|-------|----------------------------|-------|-----------------------------|-------|-------------|-------|----------------------|-------|----------------------|-------|-------|
| Metric      | MSE        | MAE          | MSE          | MAE   | MSE     | MAE   | MSE         | MAE   | MSE            | MAE   | MSE                        | MAE   | MSE                         | MAE   | MSE         | MAE   | MSE                  | MAE   | MSE                  | MAE   |       |
| Electricity | 96         | <b>0.138</b> | <b>0.223</b> | 0.155 | 0.247   | 0.155 | 0.247       | 0.175 | 0.267          | 0.151 | 0.245                      | 0.161 | 0.251                       | 0.157 | 0.250       | 0.166 | 0.256                | 0.160 | 0.251                | 0.171 | 0.259 |
|             | 192        | <b>0.160</b> | <b>0.258</b> | 0.170 | 0.259   | 0.169 | 0.259       | 0.182 | 0.275          | 0.163 | 0.254                      | 0.171 | 0.261                       | 0.166 | 0.257       | 0.177 | 0.266                | 0.167 | 0.259                | 0.176 | 0.264 |
|             | 336        | <b>0.167</b> | <b>0.253</b> | 0.179 | 0.270   | 0.186 | 0.276       | 0.192 | 0.282          | 0.176 | 0.271                      | 0.188 | 0.278                       | 0.186 | 0.278       | 0.194 | 0.283                | 0.192 | 0.284                | 0.202 | 0.290 |
|             | 720        | <b>0.213</b> | <b>0.299</b> | 0.220 | 0.305   | 0.228 | 0.311       | 0.234 | 0.317          | 0.225 | 0.312                      | 0.228 | 0.312                       | 0.238 | 0.321       | 0.238 | 0.319                | 0.245 | 0.325                | 0.246 | 0.328 |
|             | Avg        | <b>0.170</b> | <b>0.258</b> | 0.181 | 0.270   | 0.185 | 0.273       | 0.196 | 0.285          | 0.179 | 0.271                      | 0.187 | 0.276                       | 0.187 | 0.277       | 0.194 | 0.281                | 0.191 | 0.280                | 0.199 | 0.285 |
| Weather     | 96         | <b>0.160</b> | <b>0.210</b> | 0.161 | 0.208   | 0.173 | 0.216       | 0.167 | 0.215          | 0.160 | 0.205                      | 0.166 | 0.210                       | 0.167 | 0.216       | 0.173 | 0.219                | 0.170 | 0.219                | 0.164 | 0.213 |
|             | 192        | <b>0.207</b> | <b>0.250</b> | 0.213 | 0.253   | 0.218 | 0.255       | 0.216 | 0.260          | 0.215 | 0.253                      | 0.214 | 0.254                       | 0.213 | 0.254       | 0.222 | 0.260                | 0.213 | 0.254                | 0.213 | 0.252 |
|             | 336        | <b>0.253</b> | <b>0.284</b> | 0.269 | 0.292   | 0.274 | 0.296       | 0.271 | 0.297          | 0.264 | 0.289                      | 0.269 | 0.294                       | 0.266 | 0.291       | 0.271 | 0.296                | 0.273 | 0.296                | 0.256 | 0.287 |
|             | 720        | <b>0.313</b> | <b>0.332</b> | 0.349 | 0.346   | 0.353 | 0.343       | 0.348 | 0.349          | 0.342 | 0.340                      | 0.346 | 0.343                       | 0.343 | 0.332       | 0.352 | 0.348                | 0.346 | 0.344                | 0.338 | 0.331 |
|             | Avg        | <b>0.233</b> | <b>0.269</b> | 0.248 | 0.275   | 0.255 | 0.278       | 0.251 | 0.280          | 0.245 | 0.272                      | 0.249 | 0.275                       | 0.247 | 0.273       | 0.255 | 0.281                | 0.251 | 0.278                | 0.243 | 0.271 |
| Traffic     | 96         | <b>0.389</b> | <b>0.259</b> | 0.479 | 0.311   | 0.486 | 0.318       | 0.484 | 0.314          | 0.443 | 0.299                      | 0.461 | 0.303                       | 0.459 | 0.308       | 0.494 | 0.301                | 0.494 | 0.320                | 0.451 | 0.317 |
|             | 192        | <b>0.405</b> | <b>0.258</b> | 0.493 | 0.316   | 0.483 | 0.314       | 0.491 | 0.313          | 0.458 | 0.300                      | 0.484 | 0.311                       | 0.473 | 0.309       | 0.495 | 0.308                | 0.487 | 0.317                | 0.486 | 0.329 |
|             | 336        | <b>0.398</b> | <b>0.286</b> | 0.503 | 0.322   | 0.490 | 0.315       | 0.503 | 0.318          | 0.471 | 0.302                      | 0.493 | 0.318                       | 0.485 | 0.311       | 0.507 | 0.316                | 0.505 | 0.326                | 0.479 | 0.317 |
|             | 720        | <b>0.437</b> | <b>0.290</b> | 0.538 | 0.341   | 0.530 | 0.337       | 0.535 | 0.334          | 0.542 | 0.349                      | 0.516 | 0.327                       | 0.516 | 0.328       | 0.544 | 0.321                | 0.531 | 0.334                | 0.535 | 0.342 |
|             | Avg        | <b>0.407</b> | <b>0.274</b> | 0.503 | 0.323   | 0.497 | 0.321       | 0.503 | 0.320          | 0.479 | 0.313                      | 0.489 | 0.315                       | 0.483 | 0.314       | 0.510 | 0.312                | 0.504 | 0.324                | 0.488 | 0.326 |
| Solar       | 96         | <b>0.187</b> | <b>0.231</b> | 0.221 | 0.260   | 0.232 | 0.278       | 0.242 | 0.290          | 0.208 | 0.255                      | 0.237 | 0.280                       | 0.240 | 0.278       | 0.241 | 0.284                | 0.226 | 0.272                | 0.234 | 0.279 |
|             | 192        | <b>0.216</b> | <b>0.261</b> | 0.241 | 0.273   | 0.264 | 0.295       | 0.295 | 0.324          | 0.248 | 0.281                      | 0.281 | 0.298                       | 0.302 | 0.316       | 0.283 | 0.303                | 0.279 | 0.304                | 0.273 | 0.303 |
|             | 336        | <b>0.223</b> | <b>0.269</b> | 0.264 | 0.290   | 0.283 | 0.306       | 0.266 | 0.302          | 0.261 | 0.289                      | 0.289 | 0.309                       | 0.301 | 0.312       | 0.289 | 0.301                | 0.276 | 0.296                | 0.305 | 0.322 |
|             | 720        | <b>0.226</b> | <b>0.271</b> | 0.267 | 0.288   | 0.276 | 0.293       | 0.282 | 0.304          | 0.254 | 0.280                      | 0.292 | 0.303                       | 0.264 | 0.289       | 0.298 | 0.306                | 0.280 | 0.299                | 0.289 | 0.306 |
|             | Avg        | <b>0.213</b> | <b>0.258</b> | 0.248 | 0.278   | 0.264 | 0.293       | 0.271 | 0.305          | 0.243 | 0.276                      | 0.275 | 0.298                       | 0.277 | 0.299       | 0.278 | 0.299                | 0.265 | 0.293                | 0.275 | 0.303 |

Table 10: Full results of ablation study. All the results are selected from 4 different prediction lengths {96, 192, 336, 720}, and the look-back length is fixed to 96 for all baselines. A lower MSE or MAE indicates a better prediction, with the best in **boldface**.